

Introduction

Learning Objectives:

- What is this book about?
- Two key concepts in market microstructure: market liquidity and price discovery
- Why do people care about market liquidity and price discovery?
- Which puzzles can market microstructure address?
- The three dimensions of liquidity

0.1 What Is This Book About?

The way securities are actually traded is far removed from the idealized picture of a frictionless and self-equilibrating market offered by the typical finance textbook. In that idealized version of the trading process, all potential participants are present on the market; these participants convey to the market orders that reflect their demand or supply of securities, and they are not affected by actions of other market participants; and an auctioneer balances the quantities demanded and supplied at a single equilibrium price that reflects a consensus view of the security's "fundamental value." Real-world markets do not work like this, for two main reasons.

First, market players are not all simultaneously present on the market. Such continuous presence would be too costly in time, attention, and access costs. At any given point in time, price formation is delegated to the limited number of market participants who happen to be present. Any temporary imbalance between buy and sell orders for a security will have to be absorbed by whoever is present, especially by professional intermediaries who specialize in "making the market." Typically, market makers and other investors will absorb order imbalances only if the price is sufficiently attractive. For instance, to absorb a spate of sell orders investors will require the inducement of a sufficiently low price. As a result, the equilibrium price actually struck at any given instant may deviate from the one that would emerge if all investors participated. These price deviations generate profit opportunities, which in turn will draw in more traders. Over time the deviations are ironed out.

Second, even the limited number of participants who are present at any instant in a real-world security market have quite diverse information about the security's fundamentals: some participants are shrewd market professionals with all the latest news and state-of-the-art pricing models at their fingertips; others do not have such up-to-date information but may try to infer it from the behavior of other participants; still others may trade for reasons that are unrelated to information, for instance a need to liquidate their holdings in order to pay their bills. As a result, the order flow is a complex mix of information and noise, and a consensus price only emerges over time, as the trading process evolves and participants interpret the actions of other traders. This is another reason why a security's actual transaction price might deviate from its fundamental value, which would be assessed by a fully informed set of investors.

This book takes these deviations of prices from fundamental values seriously. We explain why and how they emerge in the trading process, and how and why they are eventually eliminated. Fortunately we can draw on a vast body of theoretical insights and empirical findings on security price formation that has been built up in the last thirty years, forming a well-defined field of financial economics known as "market microstructure." As we shall see, the study of market microstructure illuminates two key aspects of real-world markets that are neglected by textbook asset pricing models: liquidity and price discovery.

Liquidity is the degree to which an order can be executed within a short time frame at a price close to the security's consensus value. Conversely, a price that deviates substantially from this consensus value indicates illiquidity: in an illiquid market, buy orders tend to push transaction prices up, while sell orders tend to do the opposite; in extreme cases, the deviation is so great that it is not worthwhile or feasible to trade at all, and the market freezes. In other words, in an illiquid market, the best price at which a security can be bought (ask price) is considerably above the best price at which it can be sold (bid price). And in fact the difference between these two prices—the bid-ask spread—is a common measure of illiquidity. Liquidity differs greatly among securities and over time, one of the aims of this book is to explain why this is so. For instance, Table 1 shows the bid and ask prices at which U.S. stocks issued by companies of different sizes (measured by market capitalization, that is, the market value of all their outstanding shares) could be sold or bought on January 12, 2021. Clearly, the stock of large companies such as Amazon and Boeing is extremely liquid, with bid-ask spreads in the range of a few basis points: 2 for Amazon and 4 for Boeing (a basis point is a hundredth of 0.01%). In contrast, for smaller companies, the spread is an order of magnitude larger: for instance, almost 4% for Borr Drilling—a substantial illiquidity cost for potential investors.

Liquidity also fluctuates significantly over time. During the financial crisis of 2008, the average bid-ask spreads for stocks listed on the major exchanges

Table 1 Bid and Ask Prices Quoted for Selected U.S. Stocks on January 12, 2021

Stock	Amazon	Boeing	Campbell Soup	Borr Drilling
Market capitalization (bn \$)	1566.88	117.65	13.78	0.23
Best bid price	3124.00	207.71	45.48	0.99
Best ask price	3124.75	207.79	45.36	1.03
\$ bid-ask spread	0.75	0.08	0.38	0.04
% bid-ask spread	0.02%	0.04%	0.83%	3.96%

worldwide increased dramatically, from about 3% in the first half of 2008 to 6% in the six months following the failure of Lehman Brothers in September. The average spread peaked at over 6.5% in the period of great uncertainty preceding the announcement of the Citibank rescue on November 23.¹ The connection between uncertainty and illiquidity is underscored by the fact that it was financial stocks whose spreads increased the most sharply by far during those months. The spikes in bid-ask spreads on the stock market coincided with even greater disruptions in the interbank market and the markets for credit default swaps (CDS) and many asset-backed securities. The lack of liquidity was so intense that at some points markets simply seized up. This book examines and explicates the causes of such dramatic changes in market liquidity.

Price discovery is the speed and accuracy with which transaction prices incorporate information available to market participants. Markets sometimes display an astonishing ability to locate information about recent events and extract its implications for underlying stock values. For example, in the wake of the space shuttle Challenger explosion at 11:39 a.m. EST on January 28, 1986, the stock market very quickly determined which of the four potential contracting manufacturers was at fault for the defective parts of the shuttle: within fifteen minutes, there was a sell-induced New-York Stock Exchange (NYSE) trading halt in the shares of only one company, Morton-Thiokol. By the end of the day its shares had fallen by 11.86%, while Lockheed, Martin-Marietta, and Rockwell fell by much less (Maloney and Mulherin 2003). By contrast, the general public did not learn of the cause of the crash until two weeks later, on February 11, when Nobel-winning physicist Richard Feynman demonstrated that there were problems with Morton-Thiokol's booster rockets. This episode illustrates the market's ability to create knowledge out of a multitude of individual trades, each of which manages to contribute a small piece of

¹ These numbers are drawn from Beber and Pagano (2013), who analyze daily closing bid and ask prices for 16,491 stocks listed on the exchanges of from 30 countries and present in the Datastream data base.

information to the overall picture. In this sense, “securities markets are a vehicle for amalgamating unorganized knowledge” (Maloney and Mulherin 2003, p. 474).

This episode also illustrates a general but unintuitive point that will receive considerable attention here, namely, that there is a tension between price discovery and liquidity. When price-relevant information gets to the market by means of trading pressure rather than a public announcement, liquidity suffers. In fact, just as it became apparent that the sell orders of some market participants might be based on superior information about Morton-Thiokol’s responsibility for the disaster, the market for its shares became most illiquid: the NYSE specialist (market maker) dealing in Morton-Thiokol’s stock decided to halt trading, to avoid making a market in a situation where he might very easily lose money to informed traders.

0.2 Why Should We Care?

Why is market liquidity important? Asset managers and ordinary investors care about liquidity insofar as it affects the return on their investments, simply because illiquid securities cost more to buy, and sell for less. Therefore, illiquidity eats into the return. When markets are less than perfectly liquid, investors cannot buy and sell at the same price, and the bid-ask spread is typically wider for large trades. Thus, analysis of the way liquidity arises, builds, or vanishes may be very important in evaluating the portfolio choices of an asset manager or an ordinary investor.

For the same reasons, liquidity is a key concern of all the professionals who specialize in providing securities trading services, such as the trading desks of institutional investors (mutual funds, pension funds, and hedge funds) and retail stock brokers: locating the most liquid trading venue or timing trades so as to minimize trading costs is the key to providing good-quality service.

Beside being a source of costs, the trading process can also be a source of risks for investors. Insofar as liquidity can vary over time in ways that are not perfectly predictable, it can heighten the risk engendered by the unpredictability of assets’ fundamentals. So investors will require compensation not only for the expected trading costs associated with illiquidity but also for the additional risks. For both of these reasons, illiquidity affects equilibrium prices, which must discount not only risky future cash flows generated by the asset, but also the future trading costs that its holders may incur and the associated risks. The need to compensate investors for illiquidity creates a link between the field of market microstructure and that of asset pricing, which we will explore in Chapter 10.

But if illiquidity lowers securities prices, affecting the cost of capital for the issuers, then it will also affect these companies' day-to-day decisions on capital expenditure. The global financial crisis of 2008 is a telling example of the linkage between market liquidity, asset prices and economic activity: the drying up of liquidity in several securities markets in 2008 was associated with plunging asset prices and drastic reductions in security issuance and real investment by firms. And although these drops largely represented a correction of previous overpricing and over-investment, the illiquidity of securities markets undoubtedly amplified the effects of the revision in fundamentals. This episode illustrates why policy makers take such a strong interest in the liquidity of securities markets and in how policies and regulations affect it.

Investors, issuers, and policy makers naturally care not just about liquidity but also about the speed of price discovery. This determines the amount of information that at any instant is embodied in the price of a security, and hence how reliable that price is as a reference point for managers' real investment decisions. An informationally efficient price is also useful as a benchmark for evaluating the performance of the firm's management, and for devising equity- and option-based compensation schemes that provide the proper incentives. In short, market microstructure issues prove to be relevant to corporate finance choices, on such matters as capital budgeting and management compensation, as we shall see in Chapter 12.

0.3 Some Puzzles

This book explains a number of puzzling phenomena in securities markets. Let us consider a few specific examples of issues that can be attacked and understood using the analytical tools and empirical methods of market microstructure.

1. **Why does liquidity change over time, and why does it drop at times of crisis?** As we have seen, securities markets became much more illiquid during the financial crisis of 2008, especially in the second half of the year. This also happened at the start of the COVID-19 pandemic of 2020, when the average bid-ask spread on U.S. stocks in the CRSP US Stock database roughly doubled relative to the end of 2019 (Figure 0.1).

Stock market liquidity also exhibits long-run trends over time: in the U.S. it has been basically improving since World War II, with average bid-ask spreads gradually declining from about 0.6% in the 1950s to about 0.2% around 2000, with an especially large drop in the 1990s (Jones 2002). Moreover, liquidity varies systematically over a much more limited time frame as well: within a single trading day, bid-ask spreads tend to feature

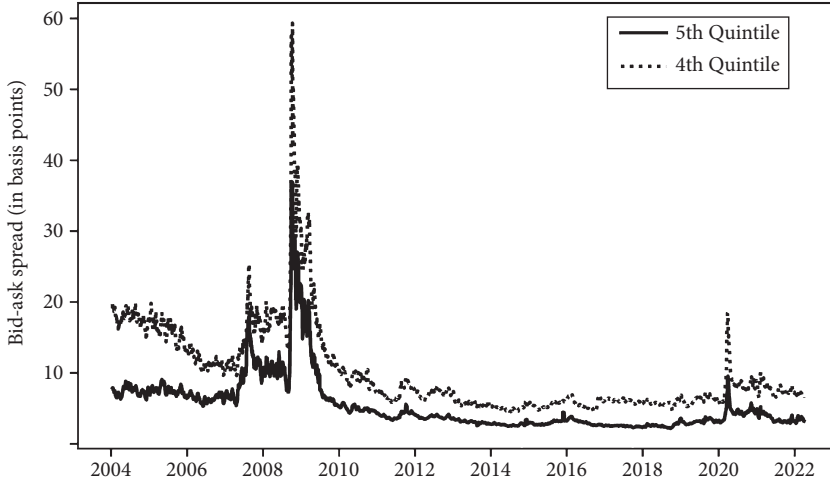


Figure 0.1 10-day rolling average closing bid-ask spread (in basis points) for AMEX, NYSE, and Nasdaq stocks by market capitalization (two first quintiles) in the CRSP data set.

Source: WRDS.

a U-shaped pattern, higher at the open and at the close than during the rest of the trading day. In addition, for individual stocks liquidity tends to drop in connection with special events, such as takeover battles, or in the wake of other dramatic price-relevant events, as exemplified by the Challenger explosion episode discussed above. The models of security price determination presented in Chapters 3 and 4 offer insight into the reasons for these low- and high-frequency empirical regularities, and Chapter 11 will focus more specifically on how market liquidity changes at times of financial instability.

2. **Why do large trades move prices up or down, and why are these price changes subsequently reversed?** One of the most widely observed patterns in securities markets is that a large and sustained flow of buy or sell orders puts temporary pressure on prices: buy orders tend to drive them up, and sell orders tend to push them down, but these price movements are partly or wholly reversed after some time. This is apparent, for instance, in the wake of redemptions of shares of equity mutual funds: to repay the investors who redeem their shares, mutual funds must in turn liquidate part of their stock holdings, and thus generate a wave of sell orders for the corresponding stocks. The average price impact associated with large mutual fund redemptions is shown in Figure 0.2, counting in months from

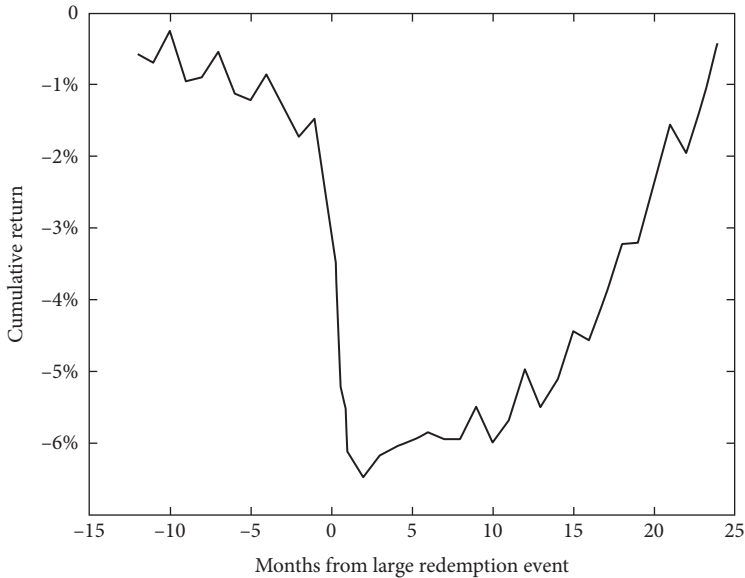


Figure 0.2 Average cumulative return of equities held by mutual funds experiencing large redemptions, counting months from the date of the large-redemption event.

Source: fig. 12 in Duffie (2010).

the date of the large-redemption event.² The figure shows that, while the prices of the stocks most affected by mutual funds' sales drop steeply in the month of the redemption, on average they completely recover in the subsequent two years. Chapter 3 will present evidence that such price patterns are not limited to mutual fund redemptions, as they are generally observed for any large security transaction, although often security prices fail to revert fully to the initial level afterwards. The chapter will provide a framework to explain these price responses to large orders.

- 3. Why has securities trading become more fragmented?** Practitioners like to say, "Liquidity begets liquidity," meaning that people like to trade at the same time of day and in the same venue as many other market participants. This suggests that security trading should tend to gravitate towards a single trading platform, which would therefore offer greater liquidity to investors than a market fragmented across several platforms. In light of this, it may

² This figure is from Duffie (2010). Based on data for the net outflows from each mutual fund, Duffie (2010) computes total outflows across mutual funds for each stock, normalized by trading volume, so as to determine price pressure indices for each equity, and then plots the cumulative returns of equities in the top decile according to price pressure, starting from the redemption date.

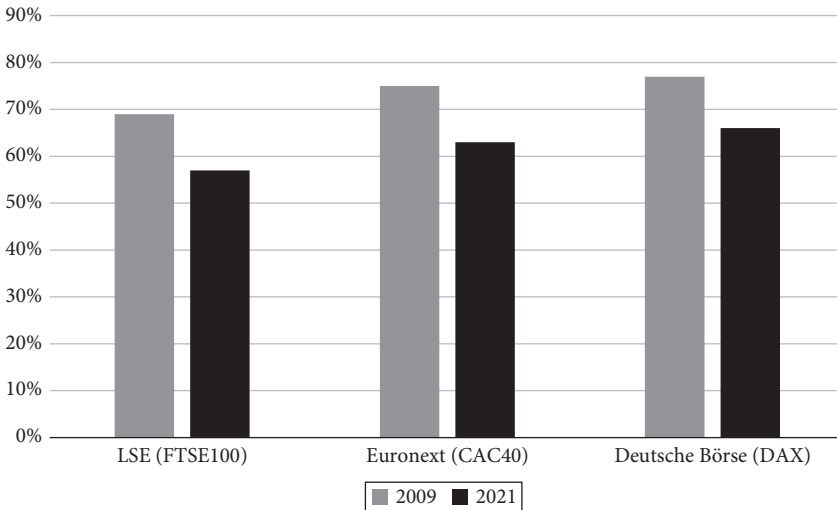


Figure 0.3 Share of trading volume in primary exchange for European stocks.

Source: Cboe Global Markets website (<https://www.cboe.com/europe/equities/>) and authors' calculations.

appear puzzling that trading in equity markets has become increasingly fragmented over time over the last two decades.

This is illustrated by Figure 0.3, which shows that between 2009 and 2021 the share of trades executed in the primary exchange for a stock (that is, the exchange on which the stock is listed) dropped substantially for stocks belonging to the main European stock indices. This is due to the entry of new trading platforms, which have eroded the share of the main regulated exchanges in the respective stock markets, namely, the share of the London Stock Exchange (LSE) for stocks belonging to the FTSE100 index, Euronext for French stocks included in the CAC40 index, and Deutsche Börse for German stocks included in the Dax index. An even more dramatic drop occurred in the United States, where the NYSE share of total trading volume in NYSE-listed shares dropped from 79.1% in January 2005 to 25.1% in October 2009 (SEC 2010). This drop in the share of trades executed in primary exchanges is part of a long-run trend: at the beginning of the new millennium, most of stock trading volume was concentrated in the respective domestic primary markets. Chapter 7 inquires into the reasons for this increase in the fragmentation of security markets, and its implications for market liquidity and for price discovery.

4. **Why do some traders seek to hide their intended trades, and others to disclose them?** Some traders really keep their cards close to the chest, submitting their orders in a way that does not reveal their true

size. For example, as Chapter 2 demonstrates, they may submit “hidden orders.” Alternatively, they may go off-exchange altogether and trade on “dark pools,” where orders are not displayed. Indeed, in the E.U. 45% of equity trading occurs outside regulated markets, hence on platforms that are typically less transparent (ESMA 2021a), while in the U.S. 35% of equity trading occurs off-exchange 2021 (SEC 2020). Conversely, some traders appear to seek transparency: for, instance, some of them opt for “sunshine trading,” that is, preannounce their trading intentions. For instance, some exchange-traded funds (ETFs) follow trading rules that make their intended trades predictable, as they periodically rebalance their portfolios in order to track public indices. Bessembinder et al. (2016), who study the trades undertaken by a large exchange-traded crude oil ETF, find that both market liquidity and resilience increase significantly around this fund’s large and predictable “roll” trade dates. Chapter 8 analyzes market transparency, explaining why it harms some traders and benefits others.

5. **Why are there temporary deviations from arbitrage prices?** The absence of arbitrage opportunities is a central tenet of asset-pricing theory: assets that generate identical cash flows must command the same market price, so that there is no opportunity for profitable arbitrage trading. Nevertheless, there are instances in which the no-arbitrage condition breaks down for non-negligible periods of time. For instance, Deville and Riva (2007) use intraday transaction data to study why it takes time for the French index options market to return to no-arbitrage values after deviating from put-call parity. Similarly, de Jong, Rosenthal, and Van Dijk (2009) document deviations from theoretical price parity in a sample of twelve dual-listed companies, sometimes known as “Siamese twins.” These are pairs of companies incorporated in different countries that contractually agree to operate their businesses as a single enterprise, while retaining their separate legal identities and existing stock exchange listings, as in the case of Royal Dutch/Shell. These companies should trade at the same price, yet from 1980–2002 their prices actually differed so much that simple trading rules could produce abnormal returns of nearly 10% per annum in some cases, after adjusting for transaction costs and margin requirements. In Chapter 11 we explore why these deviations can persist, and how they relate to market liquidity.
6. **Why has high-frequency trading become prevalent in stock markets?** In the past two decades, trading has become increasingly automated: orders are more and more frequently generated by computer algorithms, automatically routed to electronic trading platforms to be executed, and often immediately cancelled after being placed on the platform. Such “high-frequency trading,” which requires huge investments in technology,

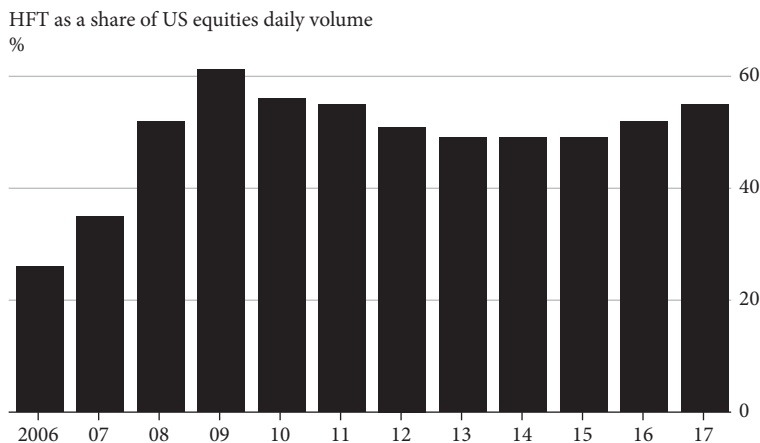


Figure 0.4 Market share of U.S. stock trading volume performed by high-frequency traders.

Source: “How High-Frequency Trading Hit a Speed Bump,” *Financial Times*, January 1, 2018.

telecommunication networks, and human capital by market participants, has become prevalent in stock markets. This is illustrated by Figure 0.4: the fraction of high-frequency trading in U.S. stocks rose from less than 30% in 2006 to over 60% in 2009, and then leveled off around 50% in subsequent years. ESMA (2021b) estimates high-frequency trading in E.U. shares to be about 60% of the total, and algorithmic non-high-frequency trading to account for another 15%. Chapter 9 of this book explores what can account for this drastic change in trading technology, and which effects it can be expected to produce on market liquidity, price discovery, and volatility.

0.4 Three Concepts of Liquidity

0.4.1 Market Liquidity

In this book we use the word *liquidity* to indicate the ability to trade a security quickly at a price close to its consensus value, that is, in the sense of “market liquidity.” But readers need to be aware that this is only one of three commonly used definitions of liquidity. As they are closely related, these three distinct concepts are often referred to interchangeably in the context of the same discourse in financial press articles and policy discussions. This can be confusing, so it is useful to clarify what liquidity means outside the context of security trading.

0.4.2 Funding Liquidity

When referring to banks or companies, liquidity is generally taken to mean having sufficient cash or the ability to obtain credit at acceptable terms, to meet obligations without incurring large losses. We can refer to this notion as “funding liquidity.” Maintaining adequate funding liquidity is particularly important for banks, which typically engage in maturity transformation. That is, they use short-term liabilities (bank deposits or repurchase agreements) to fund long-term assets (loans to companies and households). Hence, to be able to satisfy the claims of their depositors or creditors, they must maintain an adequate buffer of cash and short-term assets that can be readily liquidated. Banks have other options for generating liquidity, such as selling loans, borrowing from other banks, or borrowing from a central bank such as the U.S. Federal Reserve or the European Central Bank. However, it could still happen that, say due to a loss of confidence in the bank, depositors may wish to withdraw funds in excess of the bank’s cash reserves plus the amount it can raise by selling short-term assets (such as treasury bills or commercial paper) or obtaining overnight credit on the interbank market. Such a “bank run” is described, in fact, as a liquidity crisis, and in the absence of sufficient liquidity provision by the central bank, the distressed bank will be driven into bankruptcy and forced to liquidate its loan portfolio.

Funding liquidity is related to market liquidity in several ways. First, both have value for the same reason: people want to hold assets that can be immediately transformed into consumption, as for instance when the owner suffers a shock (e.g., a health problem or loss of job) or discovers an unforeseen opportunity (e.g., a very cheaply priced house or a very attractive business project). Since it is hard to insure against such individual-specific liquidity shocks, people try to self-insure by holding demand deposits, which they can withdraw without notice in case of need. That is, they prize funding liquidity. By the same token, investors value market liquidity, that is, they prefer assets that can be sold quickly in case of need at prices not far from their fundamental value.

Second, funding liquidity is itself a prerequisite for market liquidity. For instance, market makers often need access to credit to maintain a large enough inventory of the securities in which they are dealing, because they do not have enough equity. Hence, the more abundant and cheaper is the market makers’ funding liquidity, the greater is the liquidity of security markets, in the sense that investors will be able to trade securities in larger amounts at better prices. By the same token, a credit crunch—a drop in funding liquidity—may impair the liquidity of security markets, by forcing market makers to widen their bid-ask spreads and reduce their order size maximum. Chapter 11 analyzes the effect of funding liquidity on price formation in security markets.

Third, the causal relations can also be reversed. That is, market liquidity can be a prerequisite for funding liquidity, because security traders often must post margins (i.e., collateral in the form of cash or securities) to cover the risk that they may not be able to pay for the securities they are buying or deliver those they are selling. This “counterparty risk” can arise if the trader borrows in order to buy the security, or sells short (i.e., without owning it yet). However, margin requirements depend in part on the securities’ expected market liquidity: they are typically lower for securities that are expected to be more liquid and less volatile. Hence more liquid markets enable traders to fund their leveraged purchases or short sales more cheaply. This creates a feedback from market liquidity to funding liquidity.

This reciprocal feedback between market and funding liquidity becomes particularly important in times of crisis, when it can lead to liquidity spirals, with market liquidity suddenly drying up for many securities at once, as shown by Brunnermeier and Pedersen (2009). It is important to realize that, however deep and strong the relationship between them, market liquidity and funding liquidity are different concepts, and are accordingly affected by different policy actions: market liquidity by security market regulation and funding liquidity by banking regulation, specifically by the role of the central bank as “lender of last resort.” This brings us to a third possible meaning of liquidity, that is, the monetary dimension.

0.4.3 Monetary Liquidity

If we rank assets by market liquidity, the most liquid is obviously cash, which by definition is universally accepted in exchange for goods at very stable terms (except in times of hyperinflation). At intermediate levels of liquidity are financial securities such as bonds and stocks, while at the opposite extreme is real estate, which is so heterogeneous that sale typically requires considerable time and effort, or else a large price concession in exchange for quick sale.

This explains why in practice liquidity is often identified with money itself, whether defined as the cash held by households and firms and bank reserves (“monetary base”), or as broader monetary aggregates that also include bank deposits of various types (M1, M2, or M3). Especially in macroeconomics, this notion of “monetary liquidity” is prevalent. This notion of liquidity also bears some relationship to the previous two: expansion of the money supply by the central bank (say, via open market purchases of bonds or “quantitative easing”) increases the supply of funds to banks and thus tends to increase funding liquidity, and with it market liquidity, as we have seen. By the same token, a monetary contraction can be expected to reduce both funding and

market liquidity. There is a vast literature that analyzes and documents the link between monetary policy and funding liquidity (see Bernanke and Gertler 1995). Expansionary monetary policy may increase banks' loan supply either directly (bank lending channel) or indirectly by improving borrowers' net worth and thereby their borrowing capacity (balance-sheet channel). Monetary policy has also been shown to affect the liquidity of securities markets: at times of crisis, monetary expansion is associated with greater liquidity in both stock and bond markets, and bond market liquidity is forecast by money flows to government bond funds (Chordia, Sarkar, and Subrahmanyam 2005).

Of course, these relationships are neither mechanical nor stable over time, because banks and other financial intermediaries can generate different amounts of funding liquidity in the presence of the same level of money supply. And conversely, they may respond to an expansion of the monetary base by increasing their reserves with the central bank rather than by increasing their lending.

